RESEARCH PRESENTATION

# Design and Implementation of a Lightweight Multi-Layered Hybrid Intrusion Detection System for VANETs using RF-SVM Fusion

*A Dual-Layered Approach for Securing Vehicular Networks*


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Vehicular Network Security · Machine Learning · Intrusion Detection

# Introduction & Problem Context



### What are VANETs?

Vehicular Ad-hoc Networks enable real-time V2V and V2I communication to enhance road safety, traffic efficiency, and emergency response coordination.

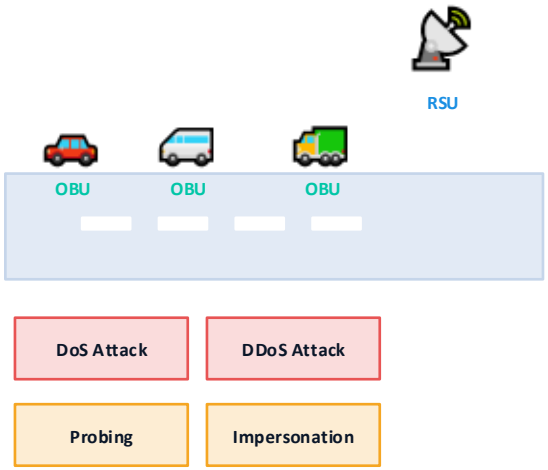
### The Attack Surface

Open wireless medium and highly dynamic topology make VANETs vulnerable to DoS, DDoS, probing, and impersonation attacks — often with life-threatening consequences.

### The Core Problem

Existing deep learning IDS impose prohibitive computational overhead incompatible with resource-constrained On-Board Units (OBUs) and Road-Side Units (RSUs).

**VANET Threat Landscape**




*Key threat vectors targeting VANET nodes*

**Deep Learning IDS**
**Too heavy for OBU/RSU hardware**

# Proposed Solution — The Hybrid Architecture

03

A lightweight, multi-layered hybrid IDS fusing Host-Based and Network-Based detection across four pipeline stages for comprehensive threat coverage.

## 01

### Data Acquisition

Raw VANET traffic capture & labeling

## 02

### HIDS Layer

OBU-level anomaly detection via RF

## 03

### NIDS Layer

RSU-level traffic analysis via SVM

## 04

### Hybrid Fusion

OR-gate decision combination layer

✅ HIDS (Random Forest) + NIDS (SVM) → Comprehensive coverage at minimal computational cost

# Host-Based IDS (HIDS) — Vehicle-Level Detection



### 🖥️ Deployment Level

Runs locally on each OBU (On-Board Unit), detecting anomalies from that vehicle's own activity and communication logs.

### 🌲 Algorithm: Random Forest

An ensemble of 100 decision trees voting on the final classification. Chosen for exceptional speed at inference time and high interpretability.

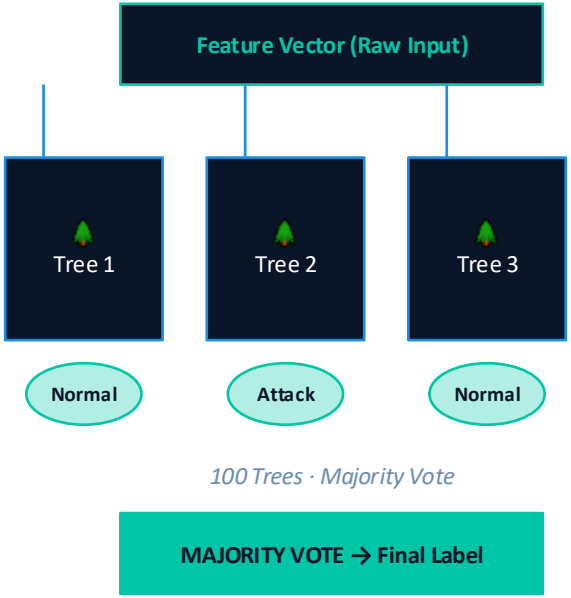
### 🛡️ Robustness

Resistant to overfitting due to bagging (bootstrap aggregation). Handles missing feature values gracefully — critical in noisy VANET environments.

### ⚡ Explainability

Feature importance scores enable human-auditable detection logic, supporting trust and regulatory compliance in safety-critical systems.

## Random Forest Architecture




📊 n_estimators = 100 | criterion = gini | bootstrap = True

# Network-Based IDS (NIDS) — RSU-Level Detection



### Deployment Level

Operates at the RSU (Road-Side Unit), aggregating and analyzing traffic flows from multiple vehicles to detect distributed, network-wide attacks.

### Algorithm: SVM with RBF Kernel

Support Vector Machine maps data into a high-dimensional feature space using a Radial Basis Function kernel, then finds an optimal separating hyperplane.

### Generalization to Unseen Attacks

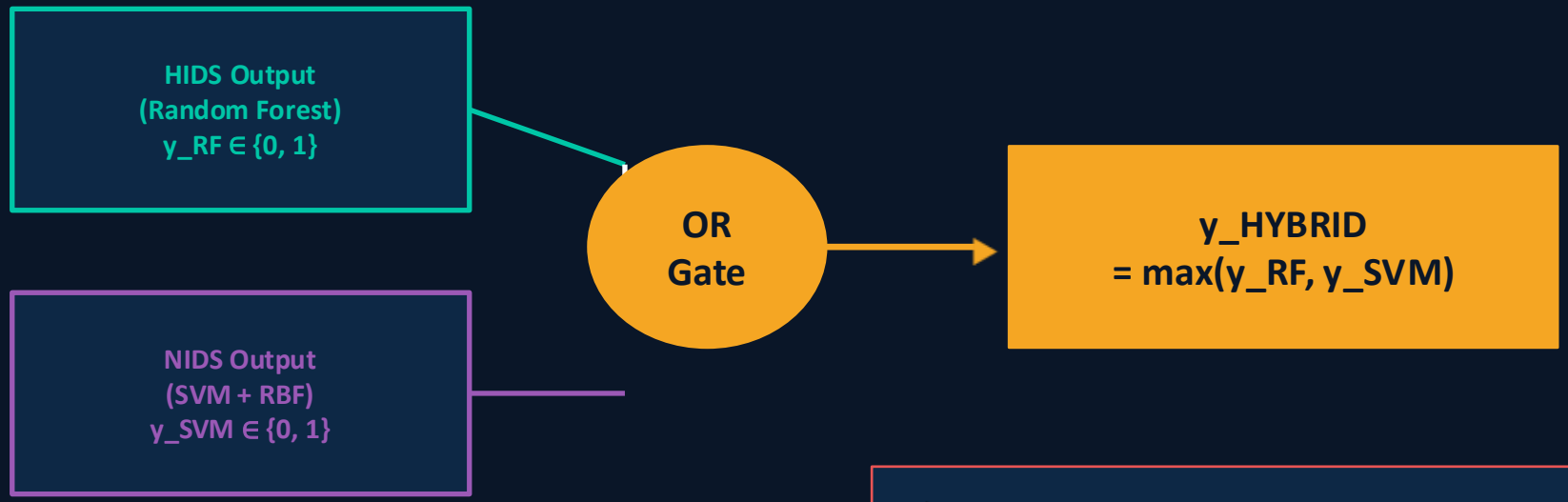
The maximum-margin classifier is specifically designed to generalize to new, unseen attack patterns — critical for zero-day threat scenarios.

### High-Dimensional Efficacy

SVM excels in high-dimensional spaces where the number of features may exceed the number of training samples — a common property of network flow data.

# The Hybrid Decision Fusion Layer






**OR-Gate Truth Table**

| y_RF | y_SVM | y_HYBRID |
|---|---|---|
| 0 (Normal) | 0 (Normal) | 0 — Normal |
| 1 (Attack) | 0 (Normal) | 1 — ATTACK |
| 0 (Normal) | 1 (Attack) | 1 — ATTACK |
| 1 (Attack) | 1 (Attack) | 1 — ATTACK |

### ⚠ Safety-Critical Design Rationale

In VANET security, a missed attack (False Negative) is far more dangerous than a spurious alert (False Positive).

**The OR-gate maximizes Recall** — ensuring that if either detector flags an anomaly, the entire system raises an alarm.

# Attack Modelling & Threat Coverage



## 💥 DoS Attack

**Primary: HIDS (RF)**

Targeted resource depletion against individual vehicles or RSUs. Floods the target with excessive requests, rendering communication unavailable.

🔎 *Local resource monitoring (CPU, bandwidth) at OBU level flags the sudden spike in load.*

## 🌊 DDoS Attack

**Primary: NIDS (SVM)**

Distributed flooding coordinated across multiple compromised vehicles to overwhelm RSUs or network infrastructure.

🔎 *Aggregated traffic analysis at RSU level reveals correlated flooding patterns invisible at the vehicle level.*

## 🔍 Probe Attack

**Both Layers**

Reconnaissance scanning to map VANET topology, enumerate active nodes, and identify exploitable services prior to a targeted attack.

🔎 *Anomalous sweep patterns detected locally by HIDS; coordinated probing detected globally by NIDS.*

## 🎭 Impersonation (Sybil)

**Indirect (HIDS)**

A single malicious node creates multiple fake identities to manipulate routing decisions, voting protocols, or traffic advisories.

🔎 *Detected indirectly via behavioral anomalies — deviations from established historical communication baselines per node.*

✅ **All four attack classes are covered · Dual-layer architecture provides no single point of detection failure**

# Experimental Setup & Dataset



## CIC-IDS2017 Benchmark Dataset

The Canadian Institute for Cybersecurity (CIC) IDS 2017 dataset provides realistic, labeled network traffic flows covering benign and attack scenarios. **It is the de facto benchmark** for IDS evaluation, encompassing DoS, DDoS, Probe, and Web attacks across multiple days of captured traffic.

## Data Preprocessing Pipeline

1. **Missing Value Handling**
   Median substitution preserves feature distributions without introducing bias from mean imputation.
2. **Feature Normalization**
   Zero-mean, unit-variance (StandardScaler) normalization ensures SVM's distance-based kernel is not skewed by feature magnitude.
3. **Label Encoding**
   Multi-class attack labels encoded for binary (attack/normal) and multi-class classification tasks.

## Evaluation Metrics

**Accuracy**

$(TP+TN)/(TP+TN+FP+FN)$

**Precision**

$TP / (TP + FP)$

**Recall**

$TP / (TP + FN)$

**F1-Score**

$2 \times (P \times R) / (P + R)$

# Comparative Evaluation Plan



Three configurations evaluated on the same CIC-IDS2017 dataset to isolate the contribution of each component:

## RF Only

**Baseline A**

Random Forest classifier alone. Vehicle-level detection only — no network-wide visibility. Serves as the HIDS standalone baseline.

- ✅ Fast inference, low overhead
- ⚠️ Misses distributed DDoS patterns

## SVM Only

**Baseline B**

Support Vector Machine alone with RBF kernel. Network-level detection only — no per-vehicle anomaly granularity. Serves as the NIDS standalone baseline.

- ✅ Global visibility, good generalization
- ⚠️ Higher inference cost, lacks local context

## RF + SVM Hybrid

**Proposed System**

Fusion of both classifiers via OR-gate decision logic. Full dual-layer coverage combining vehicle-level and network-level detection.

- ✅ Maximum Recall, comprehensive coverage
- ⚠️ Slight Precision trade-off is acceptable

📈 **Expected: Hybrid IDS achieves highest Recall & F1-Score — a practical, scalable, and safety-optimal solution for VANETs**

# Limitations & Future Work



## Current Limitations

### Static Dataset Reliance

Evaluated on CIC-IDS2017 — a static, pre-captured dataset. Does not reflect the real-time dynamic topology and mobility patterns of live VANETs.

### Indirect Impersonation Detection

Sybil/impersonation attacks are detected indirectly via behavioral anomalies only. No cryptographic identity verification is currently implemented.

### Precision Trade-off

The OR-gate logic prioritizes Recall (minimizing missed attacks) at the cost of elevated False Positive rate — potentially causing alert fatigue in low-risk scenarios.

## Future Work

### SUMO + NS-3 Integration

Deploy the hybrid IDS within SUMO (traffic simulation) and NS-3 (network simulation) to validate performance under realistic, dynamic VANET mobility models.

### Probabilistic Fusion Strategies

Replace the simple OR-gate with Bayesian or confidence-weighted fusion, enabling context-aware decision-making that dynamically balances Precision vs. Recall.

### Cryptographic Identity Verification

Integrate PKI-based certificate validation or pseudonym-based schemes to provide explicit, cryptographically verifiable Sybil attack detection.

*This research establishes a practical, computationally lightweight IDS foundation for VANET security — extensible toward real-world deployment.*